SOFTWARE ENGINEERING

LAB ASSIGNMENT 03

UML Class Diagram Design

Digital Wallet Management System (DWMS)

SUBMITTED BY — TEAM 5

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1. Introduction

A UML Class Diagram is a structural model that describes the static organization of an object-oriented system. It identifies classes, their attributes and operations, and the relationships among the classes. It is useful during analysis and design because it provides a blueprint for implementation and clarifies ownership of data and responsibilities.

1.1 Objectives

- Identify the principal domain classes implied by the Digital Wallet Management System SRS.
- Define representative attributes and methods for each class.
- Show associations and multiplicities that support wallet ownership, transactions, notifications, payment processing, and auditability.
- Keep the model aligned with the SRS while leaving unspecified implementation details open.

2. Project Scope and Class Identification

The SRS describes registration and authentication, wallet management, money transfers, payment processing, transaction history, notifications, and administration. The diagram models these responsibilities through User, Wallet, Transaction, Administrator, Notification, PaymentGateway, and AuditLog. UUIDs, enums, and service methods are design-level representations; exact database vendors and API products remain unspecified in the SRS.

SRS Reference	Design interpretation
REQ-REG-1 / REQ-AUTH-1	User registers and authenticates before wallet access.
REQ-WAL-1 to REQ-WAL-5	Wallet stores balance and supports controlled funding.
REQ-XFER-1 to REQ-XFER-5	Transactions validate recipients, enforce sufficient balance, and update both wallets atomically.
REQ-PAY-1 to REQ-PAY-5	Payment processing integrates with an external gateway and records outcomes.
REQ-NOTIF-1 to REQ-NOTIF-3	Notifications communicate transaction and account events.
REQ-ADMIN-1 to REQ-ADMIN-4	Administrator reviews users/transactions and all administrative actions are audited.

3. UML Class Diagram

The diagram presents the main domain classes and their responsibilities. A User owns a Wallet; a Wallet is associated with zero or more Transactions. Transactions may use a PaymentGateway, create notifications, and be recorded in an AuditLog. Administrator actions are represented separately because they require elevated authorization.

UML DIAGRAM

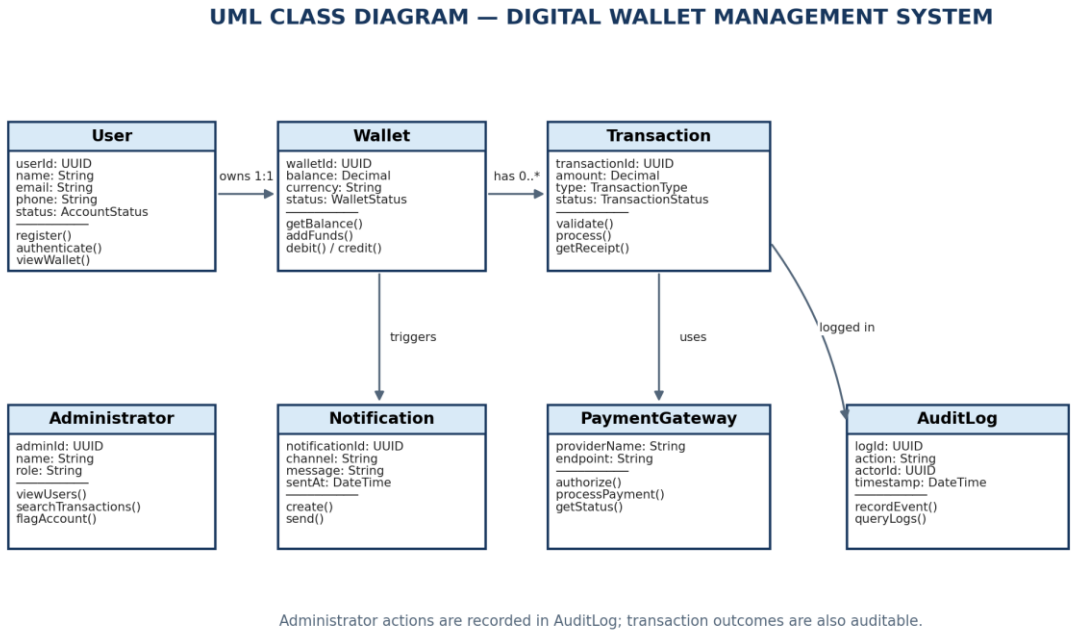


Figure 1. UML class diagram for the Digital Wallet Management System.

4. Class and Relationship Notes

- User: stores identity/contact information and exposes registration, authentication, and wallet/history access operations.
- Wallet: maintains the current balance and provides balance, credit, debit, and funding operations. Financial changes must preserve consistency.
- Transaction: captures amount, type, status, and timestamp for funding, transfers, and payments.
- PaymentGateway: abstracts external authorization and processing; DWMS consumes its interface rather than implementing the external service.
- Notification: represents delivery of account or transaction messages through a configured channel.
- Administrator and AuditLog: support monitoring, suspicious-activity handling, and traceability of privileged actions.

Conclusion

The class model translates the SRS features into a coherent object-oriented structure. It separates account, wallet, transaction, integration, notification, and administrative responsibilities. The model also highlights the system's central integrity requirement: a financial transaction must not leave wallet balances partially updated.